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Odd End Semester Examination December- 2024

Name of the Program: B. Tech Biotechnology

Name of the Course: MICROBIOLOGY

Time: 3 hrs

Semester: III

Course Code: **TBT 303**

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Discuss the significance of chemical composition of bacteria cell wall of Eubacteria and archaea.	CO1
(b)	Demonstrate the procedure for a Gram stain and describe the observed differences.	
(c)	Differentiate between transformation, transduction, and conjugation as gene transfer mechanisms.	
Q2	(20marks)	
(a)	Compare and contrast the traditional and molecular approaches to microbial taxonomy.	CO2
(b)	Explain how the germ theory of disease was established? Highlight the Koch Postulate.	
(c)	Name three significant contributors to the development of microbiology and their discoveries.	
Q3	(20 marks)	
(a)	Explain the role of selective media in microbial cultivation with examples.	CO3
(b)	Prove $G=1/K$	
(c)	A contaminated water sample (1ml) is provided to you. Explain the method for enumerating viable cells as CFU per ml in the given sample.	
Q4	(20 marks)	
(a)	Differentiate monophasic and Biphasic growth curve. Make a bacterial growth phase graph for bacteria growing in medium containing both glucose and lactose.	CO4
(b)	Illustrate the process of cultivating animal viruses in an embryonated egg.	
(c)	List the factors that influence's microbial growth. Explain any one.	
Q5	(20 marks)	
(a)	How the sterility conditions are managed while working in Laminar Air flow machine?	CO5
(b)	Explain any physical mode of sterilization using moist heating.	
(c)	Define the term "pure culture" and list the techniques used to isolate bacterial pure culture.	